SVKM's NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Programme:B. Tech (DATA SCIENCE) Year: I Semester: II

Academic Year: 2017-2018

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Subject:Managing Uncertainty

Date:09 May 2018

Marks:70
Time: 10.00 am to 1.00 pm
Durations: 3 (hrs)
No. of Pages : 5

Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume suitable data if necessary.**

Q1. A) Consider the following sales data:

Month	Sales
January	$200
February	$190
March	$200
April	$180
May	$170
June	$170

Month	Sales
July	$140
August	$150
September	$140
October	$120
November	$110
December	$90

I. Create a histogram of the data that might mislead people to believe that sales are generally increasing. (3 Marks)
II. Create a time series plot that might mislead people to believe that sales are generally increasing. (2 Marks)
III. Based on your answers to I) and II), what advice would you give to people who review business plans for potential investment? (2 Marks)

B) Find the missing frequency in the following distribution, for which the median is 48.25 and mode, is 44. (7 Marks)

Class Interval	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	Total
Frequency	3	7	?	?	?	20	16	4	150

Q2. A) The pass results of 50 students who took a class test is given below who took a test is given below

Marks	40	50	60	70	80	90
No. of Students	8	10	9	6	4	3

If the mean marks for all the students were 51.6, find out the mean marks of the students who failed. (3 Marks)

B) An examination was held to decide the awarding of a scholarship. The weights of various subjects were different. The marks obtained by 3 candidates (out of 100 in each subject) are given below

Subject	Weight	Students		
		A	B	C
Mathematics	4	60	57	62
Physics	3	62	61	67
Chemistry	2	55	53	60
English	1	67	77	49

I. Calculate the simple and weighted A.M. to award the scholarship. (5 Marks)
II. Compare simple and weighted A.M. and comment on which of the two is better measure of the average and why. (2 Marks)

C) The mean salary paid to 1500 employees of an organization was found to be Rs. 12,500. Later on, after disbursement of salary, it was discovered that the salary of two employees was wrongly entered as Rs 15,760 and Rs.9590. Their correct salaries were Rs. 17,760 and 8590.Calculate correct mean. (4 Marks)

3. A) In an attempt to estimate potential future demand, the National Motor Company did a study asking married couples how many cars the average energy-minded family should own in 1998. For each couple, National averaged the husband's and wife's response to get the overall couple response. The answers were then tabulated

Number of cars	0	0.5	1.0	1.5	2.0	2.5
Frequency	2	14	23	7	4	2

I. Calculate the variance and the standard deviation. (5 Marks)

II. Since the distribution is roughly bell-shaped, how many of the observations should theoretically fall between 0.5 and 1.5? Between 0 and 2? How many actually do fall in those intervals. (2 Marks)

B) Apply an appropriate measure of skewness to describe the following frequency distribution.

(7 Marks)

Age (years)	Number of Employees
Below 20	13
20 -25	29
25 – 30	46
30 - 35	60
35 - 40	112
40 - 45	94
45 – 50	45
50 and above	21

4) A) In a certain place it rains on one third of the days. The local evening newspaper attempts to predict whether or not it will rain the following day. Three quarters of rainy days and three fifths of dry days are correctly predicted by the previous evening's paper. Given that this evening's paper predicts rain, what is the probability that it will actually rain tomorrow? (4 Marks)

B) Suppose that we have data that follows a normal distribution of unknown mean μ and known variance Let the null hypothesis H0 be that $\mu = 0$. Let the alternative hypothesis Ha be that $\mu > 0$. Suppose we collect the following data: (5 Marks)

1, 2, 3, 6, −1

At a significance level of $\alpha = 0.05$, should we reject the null hypothesis?

C) A multiple-choice test contains 8 questions to each question (of which only one is correct). A student answers each question by rolling a balanced dice and checking the first answer if he gets 1 or 2, the second answer if he gets 3 or 4, and the third answer if he gets 5 or 6. To get a distinction, the student must secure at least 75 percent correct answers. If there is no negative marking, what is the probability that the students secure a distinction? (5 Marks)

Q5) A) This problem is refer to a golf team member.

Recently Luc bought a new driver (the longest-hitting club). Prior to buying this club he had carefully collected statistics on how far he was able to hit the ball with his old driver. The mean distance was 200 yards.

He has just begun using the new driver and is curious whether the average distance of his drives has gone up. He took a simple random sample of his drives using the new driver and came up with the following observations:

205 198 220 210 194 201 213 191 211 203

Based upon these observations can Luc conclude that the length of his drives has gone up, on average, with this new club? (5 Marks)

B) The number of calls coming per minute into a hotels reservation center is Poisson random variable with mean 3.

I. Find the probability that no calls come in a given 1 minute period. (2 Marks)
II. Assume that the number of calls arriving in two different minutes are independent. Find the probability that at least two calls will arrive in a given two minute period. (2 Marks)

C) Discuss Type I and Type II error and its significance (5 Marks)

Q6. A) A Gallup poll released on October 13, 2000 found that 47% of the 1052 U.S. adults surveyed classified themselves as "very happy" when given the choices of:

"very happy"
"fairly happy"
"not too happy"
Suppose that a journalist who is a pessimist took advantage of this poll to write a headline titled "Poll finds that U.S. adults who are very happy are in the minority." Is the pessimistic journalist's headline warranted at 5%? (4 Marks)

B) A brokerage survey reports that 30 percent of individual investors have used a discount broker, that is, one which does not charge the full commission. In a random sample of 9 individuals, what is the probability that

I. Exactly two of the sampled individuals have used a discount broker (2 Marks)
II. Not more than three have used a discount broker (2 Marks)
III. At least three of them have used a discount broker (1 Marks)

C) There are five machines in a factory. Of these machines, three are working properly and two are defective. Machines which are working properly produce articles each of which has independently a probability of 0.1 of being imperfect. For the defective machines this probability is 0.2. A machine is chosen at random and five articles produced by the machine are examined. What is the probability that the machine chosen is defective given that, of the five articles examined, two are imperfect and three are perfect? (5 Marks)

Q7. A) The average score of all sixth graders in school District A on a math aptitude exam is 75 with a standard deviation of 8.1. A random sample of 100 students in one school was taken. The mean score of these 100 students was 71. Does this indicate that the students of this school are significantly less skilled in their mathematical abilities than the average student in the district? (Use a 5% level of significance.) (5 Marks)

B) A psychologist was interested in exploring whether or not male and female college students have different driving behaviors. There were a number of ways that she could quantify driving behaviors. She opted to focus on the fastest speed ever driven by an individual. Therefore, the particular statistical question she framed was as follows:

Is the mean fastest speed driven by male college students different than the mean fastest speed driven by female college students?

She conducted a survey of a random $n = 34$ male college students and a random $m = 29$ female college students. Here is a descriptive summary of the results of her survey:

	Male	Female
No of observation	31	29
Mean	105.5	90.9
Sample standard deviation	20.1	12.2

Is there sufficient evidence at the $\alpha = 0.05$ level to conclude that the mean fastest speed driven by male college students differs from the mean fastest speed driven by female college students? (5 Marks)

C) How Box plot is useful in decision making? (4 Marks)

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